

A DEPTH FIRST SEARCH BASED APPROACH TO IMPROVING THE MOLECULAR PATHWAY SEARCH

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Molecular pathway analysis has become significant and gained greater value in bioinformatics due to improvements in computational hardware. Molecular pathways can be defined as a network of molecules that take place within a cell. The reactions between molecules are important as the results directly relate to health. Pathways can mathematically be represented as graphs, and analysis can be performed on graph models. However, the number of pathway maps is substantial. Thus, with this method, Molecular Pathway Search (MPS) analysis runs for several days or even weeks, depending on the computation capacity of the machines. Accuracy against the execution time is considered the main issue in such scenarios. Researchers have yet to address this problem completely with limited computational power. Most previously developed graph-related algorithms are executed sequentially, and sequential algorithms do not fit well with high-performance computing. One of the main objectives of this work was to enhance MPS with the help of parallel computing techniques. The Depth First Search (DFS) algorithm was implemented. MPS is divided into individual search tasks and performs DFS on each task in parallel. When there are many nodes (molecules) and connections (interactions) in the graph, the traditional DFS techniques need a very high computational power. We introduced a new termination condition to overcome this problem in this work. The termination condition was controlled with a special parameter which controls the number of visits for a node depending upon the required level of accuracy and the available computational power. With the introduced parameter, all the previously uncompleted executions were completed. A cluster computing system with a 1TB shared memory was used to evaluate the proposed method, and depending on the termination condition, it has reduced the computation time significantly.

Keywords: Parallel processing, Molecular pathways, Cluster computing, Graph search